Boris Kafidov

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Education

University of Toronto Toronto, ON
Honours B.Sc. in Computer Science & Mathematics, Focus in Artificial Intelligence *Sep. 2024 – Jun. 2028*

- **Selected Coursework:** Software Design, Systems Programming, Theory of Computation, Linear Algebra

Experience

Amazon Toronto, ON
Jr. Software Development Engineer (Internship) *Jun. 2026 – Present*

- Root-caused and shipped a TypeScript microfrontend and product API fix for data-selection discrepancies across three regions by correcting missing request context, preserving interfaces, and keeping all 707 unit tests green
- Authored the technical design and led a five-engineer review for a 12-marketplace TypeScript microfrontend migration, evaluating API and search-backed architectures for scalability, data ownership, and component reuse
- Developed TypeScript AWS CDK definitions for 18 CloudWatch alarms across three regions and desktop, mobile, and tablet surfaces, designed to replace six manual monitors with sustained escalation and incident routing
- Shipped TypeScript, semantic HTML, and CSS fixes for three accessibility defects spanning keyboard activation, button semantics, and high-contrast visibility, and escalated the shared-framework root cause to its owning team

University of Toronto, Wheeler Lab Toronto, ON
Software Engineer (Work Study) *Oct. 2025 – Mar. 2026*

- Extended a Python/PySide6 Linux GUI for DISCO microscopy, automating device control, tiling, z-stack capture, live acquisition, preset changes, and per-channel imaging workflows used by researchers during long microscope runs
- Added no-hardware startup paths with simulated ASI stage and camera fallbacks, letting research workflows run in demo mode for development, demonstrations, and regression testing without connected microscope hardware
- Built structured JSONL profiling for tile/channel acquisition, logging preset switches, camera timing, stage movement, waits, and per-tile duration to expose bottlenecks before runtime tuning and acquisition-loop fixes
- Optimized preset switching with batched Qt waits to reduce repeated device reconfiguration across imaging runs
- Refactored excitation control into selectable serial, HID, and auto backends with Thorlabs HID command framing

Projects

CreditCombo | *JavaScript, HTML, CSS, JSON, Web Workers, Cloudflare Workers, GA4* Feb. 2026 – Mar. 2026

- Built CreditCombo, a JavaScript rewards app modeling annual value across 151 cards, 51 programs, and 22 spend subcategories, user constraints, annual fees, reward valuations, locked-card ownership, and eligibility rules
- Implemented a cap-aware search algorithm with annualized spend, dual valuation modes, fee adjustment, overflow rerouting, merchant subcategories, and constrained combo search with locked cards and runtime exclusions
- Moved expensive card-combination search into a Web Worker and added request versioning, deep-link hydration, no-result diagnostics, and an LRU-style combo cache to keep repeated manual and guided runs responsive
- Shipped manual and guided optimizer flows, card browser, valuation guide, GA4 funnel events, shareable results, Cloudflare route rewrites, security headers, and a browser-only static launch with no build step or server runtime

APA Site Choice Transformer | *Python, PyTorch, Hugging Face, GCP, AWS, Git LFS* Sep. 2025

- Co-led a 5-person undergraduate team to 2nd place among 23 projects and 116 contestants, earning a $500 award
- Trained DNABERT-2-117M in PyTorch and Hugging Face on A100 GPUs with mixed precision, checkpointing, Git LFS, and about 800k labeled DNA/RNA windows assembled from public genomic datasets for supervised training
- Built preprocessing around PolyASite 2.0, PolyA_DB, GENCODE, and genome_kit to standardize genomic coordinates, extract sequence windows, separate validation chromosomes, and prevent train/eval leakage
- Reached 0.86 accuracy, 0.93 AUROC, 0.94 AUPRC, and 0.86 F1 on held-out chromosomes using sequence input

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, C
Frameworks/Libraries: PyTorch, Hugging Face Transformers, Node.js, scikit-learn, NumPy, Pandas, PySide6, Qt
Tools/Platforms: Git, Linux, Bash, AWS CDK, CloudWatch, GCP, Cloudflare Workers, Git LFS, GA4, Web Workers
Concepts: Data Structures & Algorithms, REST APIs, Unit Testing, Accessibility, Infrastructure as Code